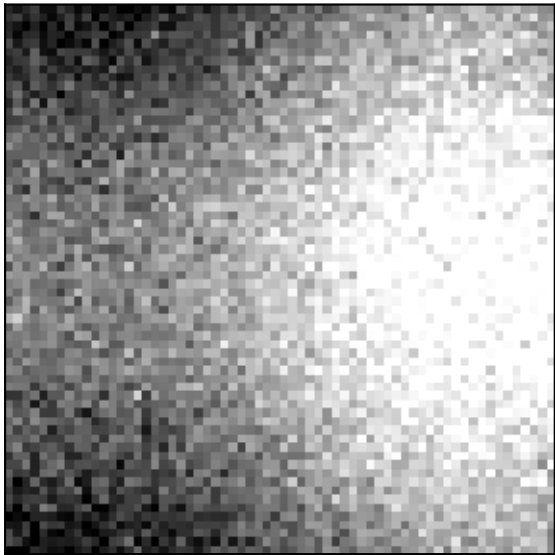


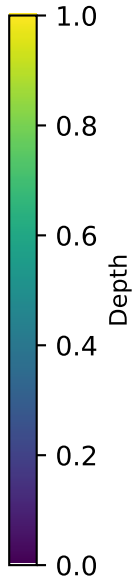
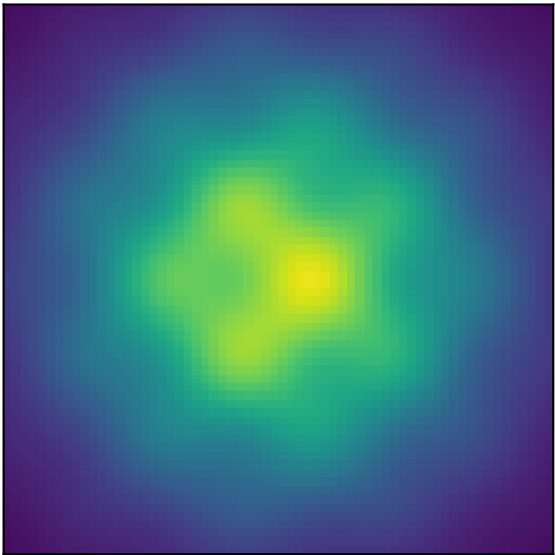
Depth-Supervised Liveness Detection: Model Learns Face Geometry

Live Face
(3D structure present)

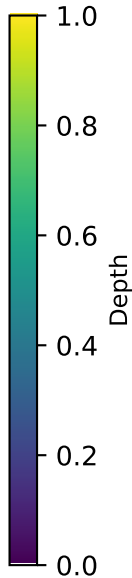
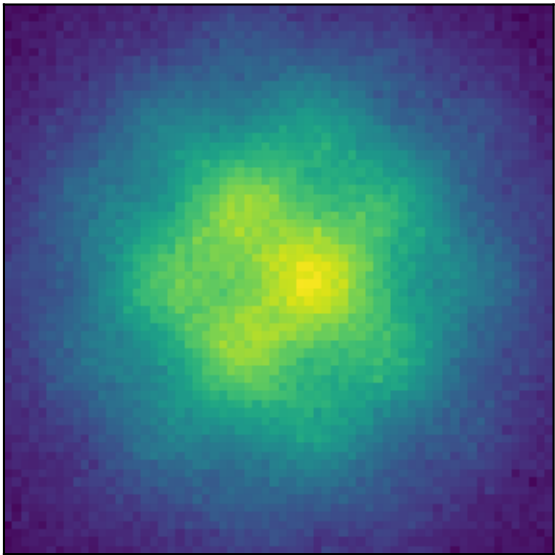
Input Image



Ground-truth Depth



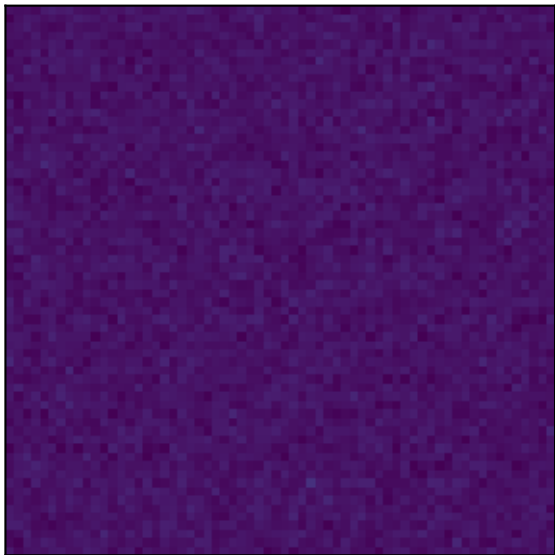
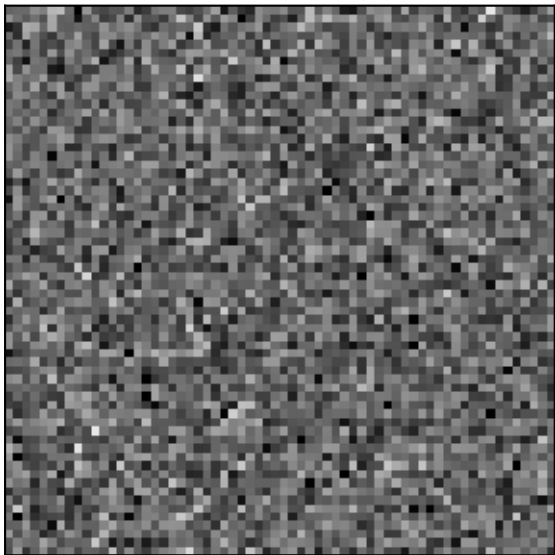
Predicted Depth



*Rich topography
(nose tip, cheeks, jaw)*

Spoof Face
(2D print, flat depth)

Input Image



*Flat profile
(uniform ~0 depth)*

